

Linux Access Control Lab Portfolio

User, Group, Permission, and ACL Administration on Linux

Prepared by	Wilson Nhlanhla Radebe
Project type	Linux System Administration lab
Repository use	GitHub Portfolio
Status	Final Portfolio

Prepared for a public GitHub repository and interview review.

1. Executive Summary

This lab demonstrates practical Linux access control in a multi-user environment. The work covers account creation, group management, directory ownership, permission hardening, ACL-based traversal access, and final access validation. The outcome is a clean departmental structure where security staff and finance staff can reach only the directories intended for them.

2. Lab Objectives

- Create standard user accounts for separate access paths.
- Create dedicated groups for security and finance operations.
- Build a shared /srv directory with departmental subdirectories.
- Apply ownership and permissions to restrict unauthorized access.
- Use setgid so new content inherits the correct group context.
- Grant minimal ACL access for traversal where needed.
- Verify the final configuration by testing access as different users.

3. Environment and Tools

Category	Details
Operating system	Linux desktop environment
Shell	bash
Utilities used	adduser, addgroup, usermod, mkdir, touch, chown, chmod, setfacl, getfacl, su, ls
Working path	/home/willyonazure/srv

4. Implementation Summary

4.1 User creation

The first step was creating the user account john with adduser. The account was initialized with a home directory, password, and standard user metadata. A second account, jane, was then created using the same process so the finance side of the lab could be tested independently.

4.2 Directory structure

A working directory named **srv** was created under the main account home path. Two departmental subdirectories were then created inside it: **security_directory** and **finance_directory**. Test files were created in each folder to represent controlled departmental content.

4.3 Groups, ownership, and permissions

Two groups were created to separate access: **security_group** and **finance_group**. The directories were assigned to root with the matching departmental group. Permissions were tightened so that only the owner and the correct group could access the directories. The setgid bit was also applied to preserve group inheritance for new content.

4.4 ACL traversal access

ACL entries were added to the parent home directory so **john** and **jane** could traverse into the lab environment without exposing broader access than necessary. This is the right model: grant only the minimum access required for the task.

4.5 Validation

The last stage tested access from **root**, **john**, and **jane**. The results confirmed the intended separation. **Root** was blocked from direct directory traversal after permissions were tightened, **john** could access the **security directory** and read **logs.txt**, and **jane** could access the **finance directory** and read **expenditure.txt**.

5. Results

- User creation completed successfully for **john** and **jane**.
- The **srv** folder structure was created and organized by department.
- Ownership and permission settings restricted each directory to the correct group.
- ACLs allowed controlled traversal without opening the environment broadly.
- Access testing proved that the final security model behaved as intended.

6. Key Takeaways

This lab is basic on the surface, but it reflects real administrative discipline. Most permission mistakes in Linux come from sloppy group design, weak ownership planning, or giving away too much access too early. The structure here avoids that problem. It separates responsibilities cleanly and proves the setup with live testing.

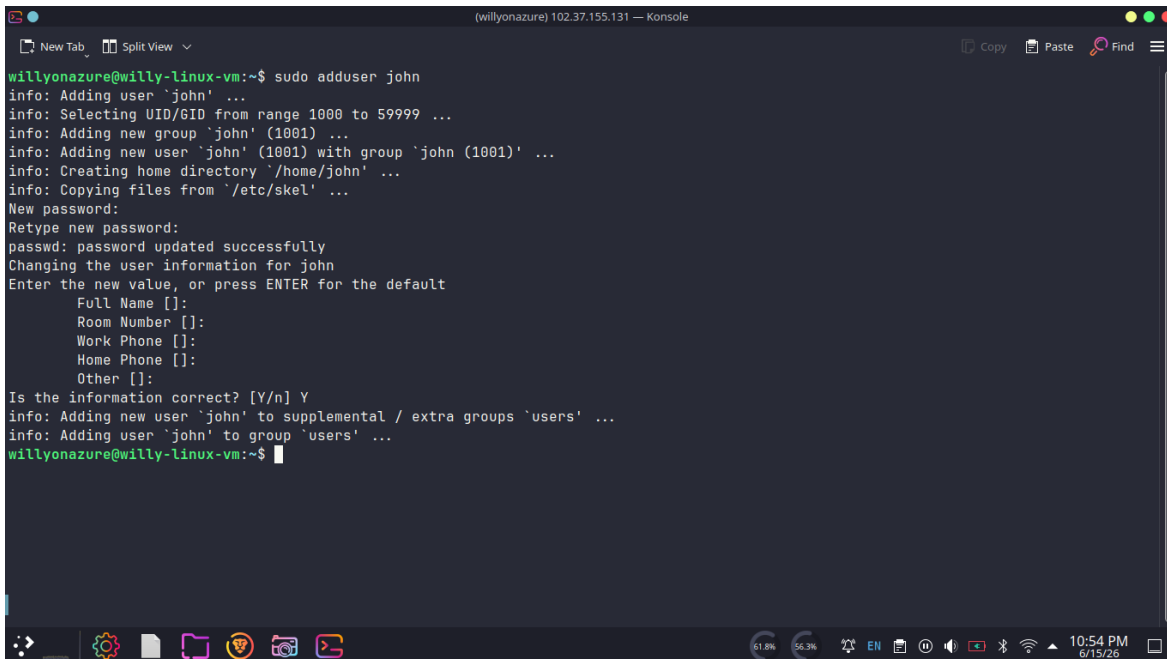
7. Conclusion

The lab was completed successfully and demonstrates the full workflow for Linux access control at a small departmental scale. The final configuration provides the right balance of structure and restriction: users can reach what they need, and nothing more.

Appendix A. Evidence Log

Each figure below is presented in chronological order. The captions match the lab sequence and are ready for GitHub publication.

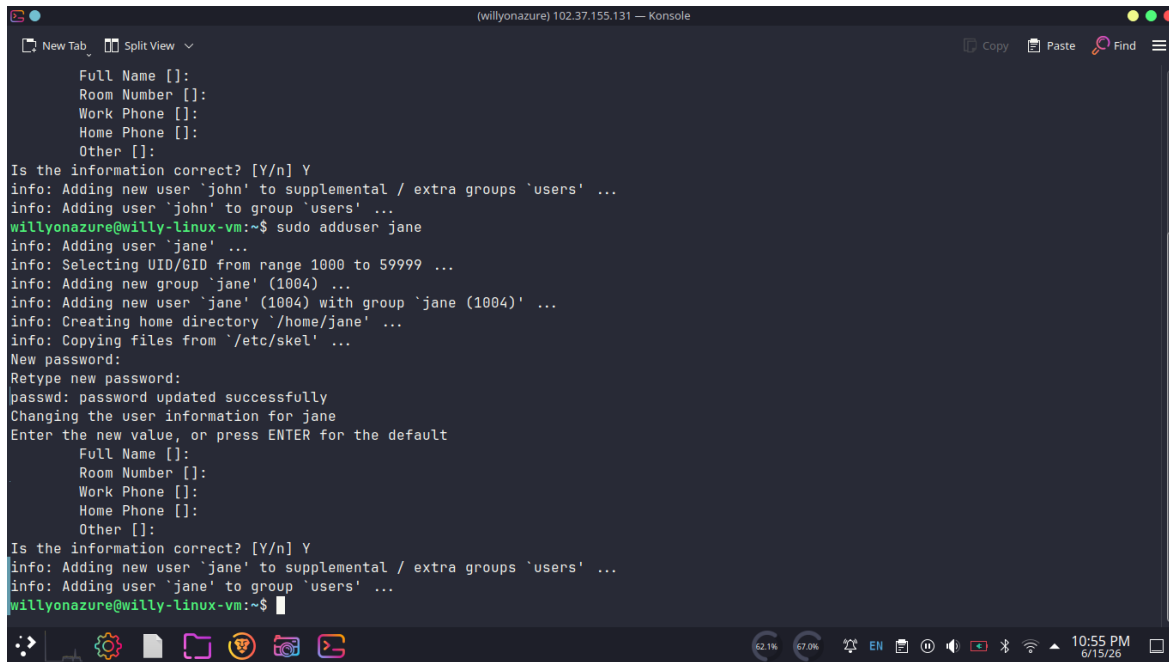
Figure 1. Creating the user account `john`.



```
willyonazure@willy-linux-vm:~$ sudo adduser john
info: Adding user `john' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `john' (1001) ...
info: Adding new user `john' (1001) with group `john (1001)' ...
info: Creating home directory `/home/john' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for john
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
info: Adding new user `john' to supplemental / extra groups `users' ...
info: Adding user `john' to group `users' ...
willyonazure@willy-linux-vm:~$
```

Source: ADDUSER JOHN.png

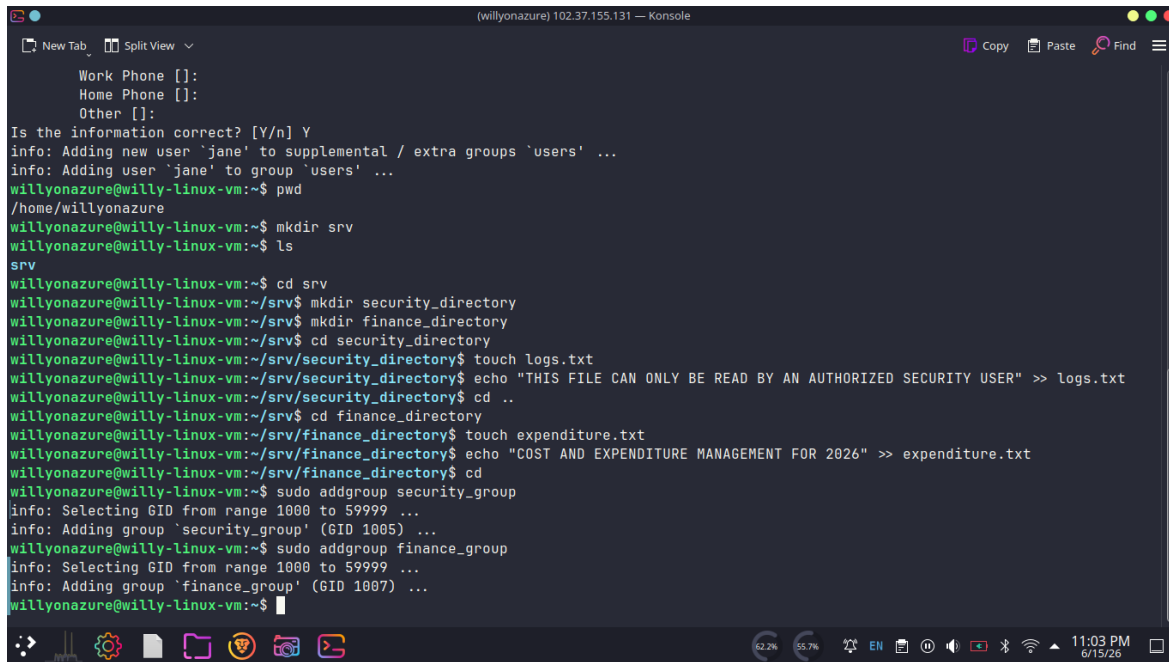
Figure 2. Creating the user account `jane`.



```
(willyonazure) 102.37.155.131 — Konsole
New Tab Split View
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] Y
info: Adding new user 'john' to supplemental / extra groups 'users' ...
info: Adding user 'john' to group 'users' ...
willyonazure@willy-linux-vm:~$ sudo adduser jane
info: Adding user 'jane' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'jane' (1004) ...
info: Adding new user 'jane' (1004) with group 'jane (1004)' ...
info: Creating home directory '/home/jane' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jane
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] Y
info: Adding new user 'jane' to supplemental / extra groups 'users' ...
info: Adding user 'jane' to group 'users' ...
willyonazure@willy-linux-vm:~$
```

Source: ADDUSER JANE.png

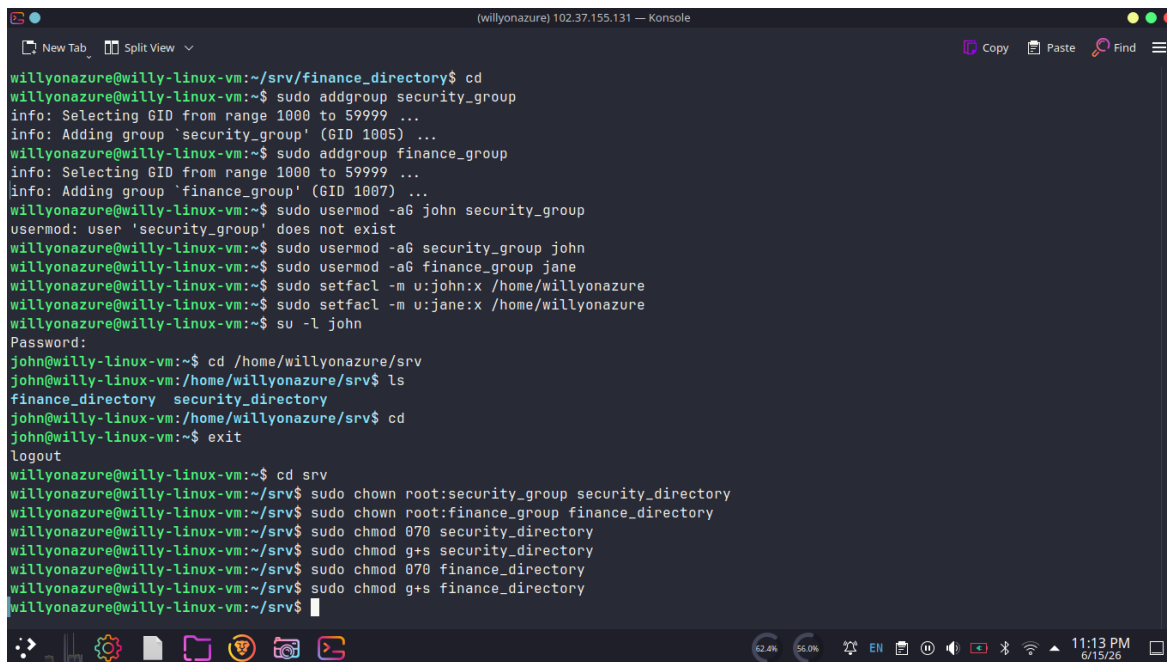
Figure 3. Building the `/home/willyonazure/srv` structure and departmental files.`



```
(willyonazure) 102.37.155.131 — Konsole
New Tab Split View
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] Y
info: Adding new user 'jane' to supplemental / extra groups 'users' ...
info: Adding user 'jane' to group 'users' ...
willyonazure@willy-linux-vm:~$ pwd
/home/willyonazure
willyonazure@willy-linux-vm:~$ mkdir srv
willyonazure@willy-linux-vm:~$ ls
srv
willyonazure@willy-linux-vm:~$ cd srv
willyonazure@willy-linux-vm:~/srv$ mkdir security_directory
willyonazure@willy-linux-vm:~/srv$ mkdir finance_directory
willyonazure@willy-linux-vm:~/srv$ cd security_directory
willyonazure@willy-linux-vm:~/srv/security_directory$ touch logs.txt
willyonazure@willy-linux-vm:~/srv/security_directory$ echo "THIS FILE CAN ONLY BE READ BY AN AUTHORIZED SECURITY USER" >> logs.txt
willyonazure@willy-linux-vm:~/srv/security_directory$ cd ..
willyonazure@willy-linux-vm:~/srv$ cd finance_directory
willyonazure@willy-linux-vm:~/srv/finance_directory$ touch expenditure.txt
willyonazure@willy-linux-vm:~/srv/finance_directory$ echo "COST AND EXPENDITURE MANAGEMENT FOR 2026" >> expenditure.txt
willyonazure@willy-linux-vm:~/srv/finance_directory$ cd
willyonazure@willy-linux-vm:~$ sudo addgroup security_group
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'security_group' (GID 1005) ...
willyonazure@willy-linux-vm:~$ sudo addgroup finance_group
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'finance_group' (GID 1007) ...
willyonazure@willy-linux-vm:~$
```

Source: MKDIR SRV AND SUBDIR.png

Figure 4. Applying ownership, group membership, and directory permissions.

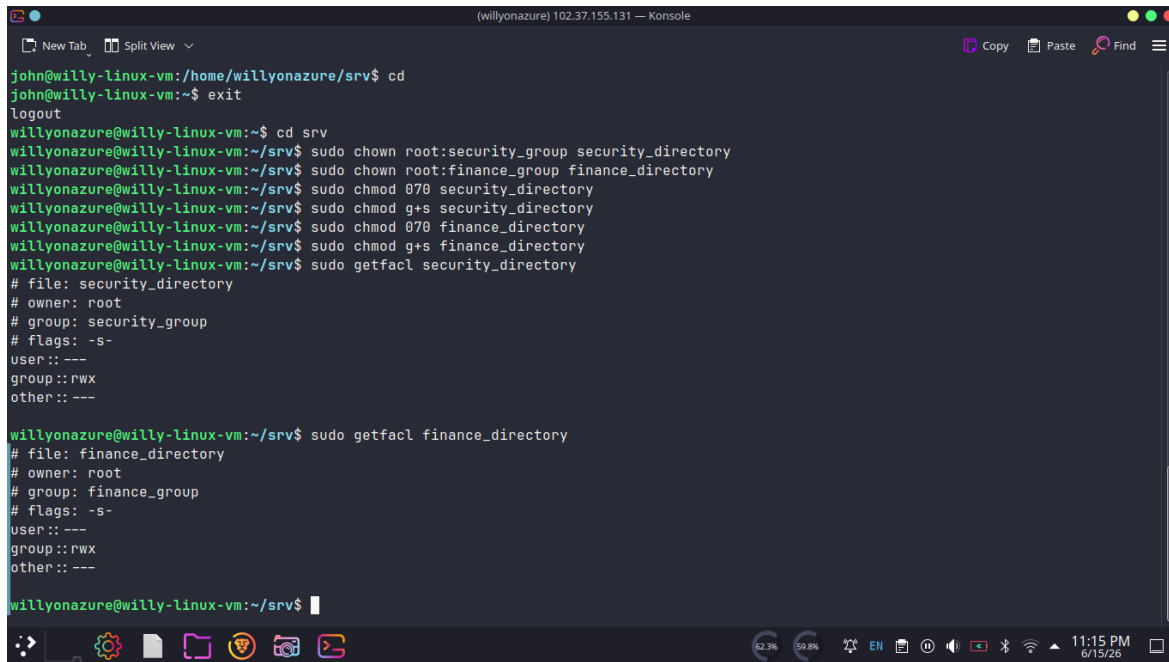


```
(willyonazure) 102.37.155.131 — Konsole
New Tab Split View
Copy Paste Find

willyonazure@willy-linux-vm:~/srv/finance_directory$ cd
willyonazure@willy-linux-vm:~$ sudo addgroup security_group
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'security_group' (GID 1005) ...
willyonazure@willy-linux-vm:~$ sudo addgroup finance_group
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'finance_group' (GID 1007) ...
willyonazure@willy-linux-vm:~$ sudo usermod -aG john security_group
usermod: user 'security_group' does not exist
willyonazure@willy-linux-vm:~$ sudo usermod -aG security_group john
willyonazure@willy-linux-vm:~$ sudo usermod -aG finance_group jane
willyonazure@willy-linux-vm:~$ sudo setfacl -m u:john:x /home/willyonazure
willyonazure@willy-linux-vm:~$ sudo setfacl -m u:jane:x /home/willyonazure
willyonazure@willy-linux-vm:~$ su -l john
Password:
john@willy-linux-vm:~$ cd /home/willyonazure/srv
john@willy-linux-vm:/home/willyonazure/srv$ ls
finance_directory security_directory
john@willy-linux-vm:/home/willyonazure/srv$ cd
john@willy-linux-vm:~$ exit
logout
willyonazure@willy-linux-vm:~$ cd srv
willyonazure@willy-linux-vm:~/srv$ sudo chown root:security_group security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chown root:finance_group finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod 070 security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod g+s security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod 070 finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod g+s finance_directory
willyonazure@willy-linux-vm:~/srv$
```

Source: *CHOWN AND CHMOD OF DIR TO GROUPS.png*

Figure 5. Verifying ACL and permission settings on the lab directories.



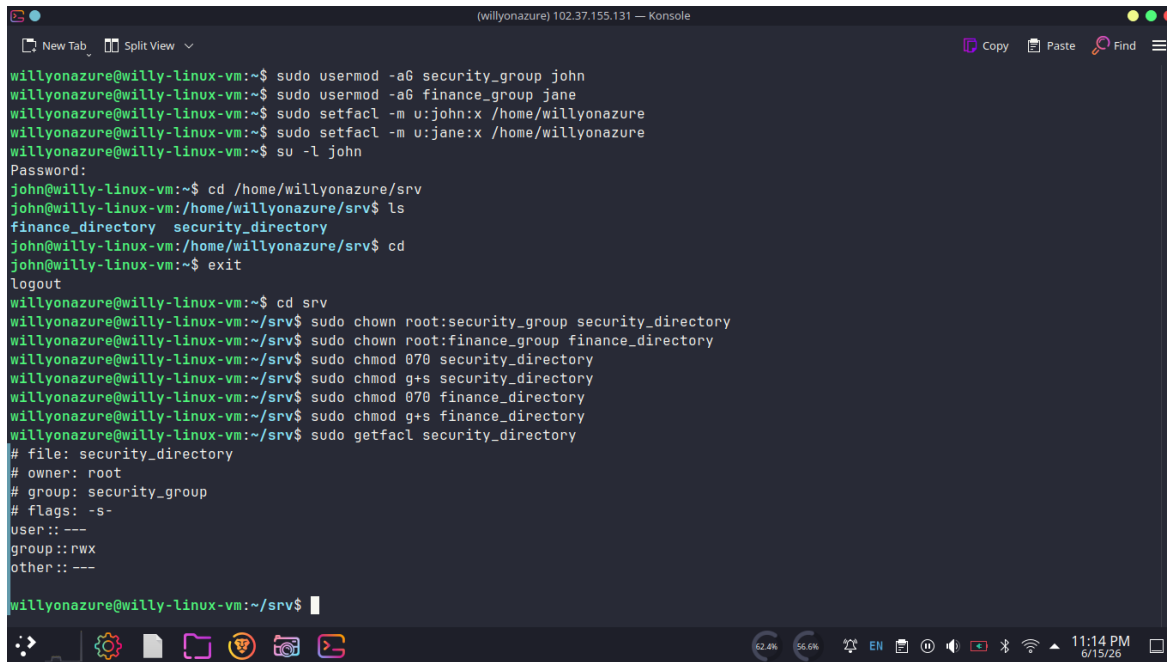
```
(willyonazure) 102.37.155.131 — Konsole
New Tab Split View
Copy Paste Find
john@willy-linux-vm:/home/willyonazure/srv$ cd
john@willy-linux-vm:~$ exit
logout
willyonazure@willy-linux-vm:~$ cd srv
willyonazure@willy-linux-vm:~/srv$ sudo chown root:security_group security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chown root:finance_group finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod 070 security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod g+s security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod 070 finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod g+s finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo getfacl security_directory
# file: security_directory
# owner: root
# group: security_group
# flags: -s-
user::---
group::rwx
other::---

willyonazure@willy-linux-vm:~/srv$ sudo getfacl finance_directory
# file: finance_directory
# owner: root
# group: finance_group
# flags: -s-
user::---
group::rwx
other::---

willyonazure@willy-linux-vm:~/srv$
```

Source: GETFACL ON DIR.png

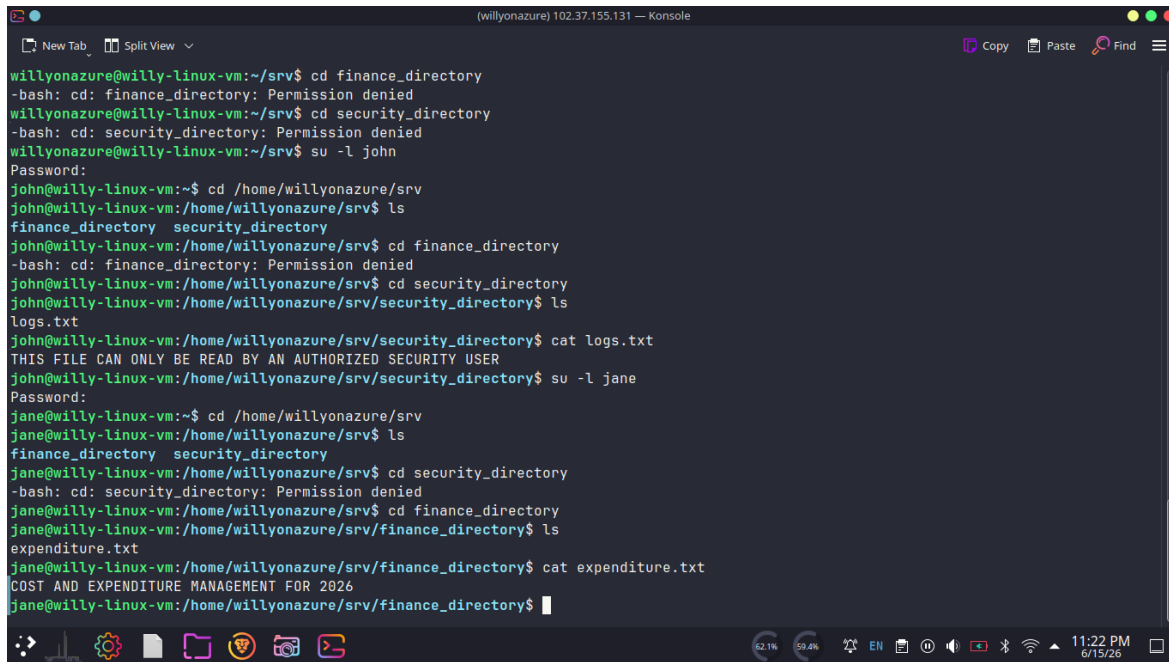
Figure 6. Additional ACL verification for `security\_directory`.

A terminal window titled "(willyonazure) 102.37.155.131 — Konsole" showing a series of commands and their outputs. The user 'willyonazure' is in a VM named 'willy-linux-vm'. The commands include setting up users 'john' and 'jane', setting file permissions, and using 'getfacl' to verify the ACL for 'security_directory'. The output shows that 'security_directory' is owned by 'root', has permissions 's-' for the owner, and 'rwx' for the group 'security_group'.

```
willyonazure@willy-linux-vm:~$ sudo usermod -aG security_group john
willyonazure@willy-linux-vm:~$ sudo usermod -aG finance_group jane
willyonazure@willy-linux-vm:~$ sudo setfacl -m u:john:x /home/willyonazure
willyonazure@willy-linux-vm:~$ sudo setfacl -m u:jane:x /home/willyonazure
willyonazure@willy-linux-vm:~$ su -l john
Password:
john@willy-linux-vm:~$ cd /home/willyonazure/srv
john@willy-linux-vm:/home/willyonazure/srv$ ls
finance_directory  security_directory
john@willy-linux-vm:/home/willyonazure/srv$ cd
john@willy-linux-vm:~$ exit
logout
willyonazure@willy-linux-vm:~$ cd srv
willyonazure@willy-linux-vm:~/srv$ sudo chown root:security_group security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chown root:finance_group finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod 070 security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod g+s security_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod 070 finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo chmod g+s finance_directory
willyonazure@willy-linux-vm:~/srv$ sudo getfacl security_directory
# file: security_directory
# owner: root
# group: security_group
# flags: -s-
user::---
group::rwx
other::---
```

Source: GETFACL ON SECURITY_DIR.png

setgid Figure 7. Testing access as root, `john`, and `jane`.



The screenshot shows a terminal window titled "(willyonazure) 102.37.155.131 — Konsole". The user "willyonazure" is in a VM named "willy-linux-vm" at the "/srv" directory. They attempt to change into "finance_directory" and "security_directory", both of which are denied permission. Then, they switch to the "john" user using "su -l john". As "john", they navigate to "/home/willyonazure/srv", list the contents (showing "finance_directory" and "security_directory"), attempt to enter "finance_directory" (denied), and successfully enter "security_directory". Inside "security_directory", they list files (showing "logs.txt") and view its contents with "cat logs.txt", which displays "THIS FILE CAN ONLY BE READ BY AN AUTHORIZED SECURITY USER". They then switch to the "jane" user using "su -l jane". As "jane", they navigate to "/home/willyonazure/srv", list the contents, attempt to enter "security_directory" (denied), successfully enter "finance_directory", list its contents (showing "expenditure.txt"), and view its contents with "cat expenditure.txt", which displays "COST AND EXPENDITURE MANAGEMENT FOR 2026".

```
willyonazure@willy-linux-vm:~/srv$ cd finance_directory
-bash: cd: finance_directory: Permission denied
willyonazure@willy-linux-vm:~/srv$ cd security_directory
-bash: cd: security_directory: Permission denied
willyonazure@willy-linux-vm:~/srv$ su -l john
Password:
john@willy-linux-vm:~$ cd /home/willyonazure/srv
john@willy-linux-vm:/home/willyonazure/srv$ ls
finance_directory  security_directory
john@willy-linux-vm:/home/willyonazure/srv$ cd finance_directory
-bash: cd: finance_directory: Permission denied
john@willy-linux-vm:/home/willyonazure/srv$ cd security_directory
john@willy-linux-vm:/home/willyonazure/srv/security_directory$ ls
logs.txt
john@willy-linux-vm:/home/willyonazure/srv/security_directory$ cat logs.txt
THIS FILE CAN ONLY BE READ BY AN AUTHORIZED SECURITY USER
john@willy-linux-vm:/home/willyonazure/srv/security_directory$ su -l jane
Password:
jane@willy-linux-vm:~$ cd /home/willyonazure/srv
jane@willy-linux-vm:/home/willyonazure/srv$ ls
finance_directory  security_directory
jane@willy-linux-vm:/home/willyonazure/srv$ cd security_directory
-bash: cd: security_directory: Permission denied
jane@willy-linux-vm:/home/willyonazure/srv$ cd finance_directory
jane@willy-linux-vm:/home/willyonazure/srv/finance_directory$ ls
expenditure.txt
jane@willy-linux-vm:/home/willyonazure/srv/finance_directory$ cat expenditure.txt
COST AND EXPENDITURE MANAGEMENT FOR 2026
jane@willy-linux-vm:/home/willyonazure/srv/finance_directory$
```

Source: TESTING PERMISSIONS FOR EVERY USER ON EVERY DIR.png